

Assignment 2 — Covariance, VaR, and Copulas

FinTech 545 — Quantitative Risk Management

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Instructions

Be verbose. Explain clearly your reasoning, methods, and results in your written work. Write clear code that is well documented. With 99% certainty, you cannot write too many code comments.

Written answers are worth 8 points. Code is worth 2 points. 10 points total.

1. When finished, respond to the question in Canvas as “done.” We will record your grade there.
2. In your code repository, create a folder called “Assignment2.”
3. In that folder, include
 - a. a document (PDF) with your responses.
 - b. All code.
 - c. A README file with instructions for us to run your code.

Data for the problems is in the CSV files alongside this document in the class repository.

This assignment covers Weeks 1 through 5, with most of the weight on Weeks 3, 4, and 5. Expect to use the earlier weeks throughout. Moments, distribution fitting, AICc, and the portfolio variance formula all come back.

Unless a problem says otherwise, VaR and ES are at $\alpha = 5\%$ and are reported as positive numbers for a loss.

How These Problems Are Graded

Every problem below is in three parts, and the middle part is the one you are most likely to skip.

1. **Predict** – look at the data, or a plot of it, and say what you expect and why. Do this before you fit anything.
2. **Fit** – run the models.
3. **Reconcile** – say whether the fit agreed with you. If it did not, say what you had missed.

Most of the written credit sits on parts 1 and 3. Correct code with no reasoning attached does not earn full marks. A wrong prediction, honestly reported and then explained, earns nearly all of them. There are no points deducted for wrong predictions. We are grading whether you understand what the numbers mean, not whether you can call a library function.

1 Correlations from Mismatched Histories (1.5 points)

problem1.csv holds 250 days of returns for four stocks, A through D, and an index, IDX. A blank cell means the asset did not trade that day. A and B share a calendar. C trades on a foreign calendar with more holidays. D listed partway through the sample. The index trades every day.

IDX is built from the four stocks with fixed weights of 0.40, 0.30, 0.20, and 0.10 on A, B, C, and D. It does not match that basket exactly, so there is a small tracking error.

Predict. Build a table of how many days each pair of series is jointly observed, and count the days on which all five trade. Then answer, before computing any correlation:

- We can build the correlation matrix from the complete rows only, or pairwise. Which of the two can return a matrix that is not PSD, and why? Which is guaranteed PSD?
- IDX is almost a linear combination of the other four. What does that imply for the smallest eigenvalue of the true correlation matrix? Using that, say how exposed this particular data set is to the failure in part (a).
- Which correlations do you expect to be the least reliable, and why?

Fit.

- Compute the complete case and the pairwise correlation matrices, and the eigenvalues of each. Attempt a Cholesky factorization of each and report what happens.
- Build a covariance matrix from the pairwise correlations and the full history standard deviation of each series. Compute the variance of the portfolio that is long one unit of IDX and short the weighted basket. Report it.
- Repair the pairwise correlation matrix with Rebonato and Jackel and with Higham. For each, report the smallest eigenvalue and the Frobenius distance from the pairwise matrix. Recompute the tracking portfolio variance from part (e).

Reconcile.

- Explain the result in part (e) in terms of the definition of positive semi-definiteness from Week 2.
- Which entries did the Higham repair move the most? Compare that with your answer to part (c). The repair was not told how many days sit behind each entry. What is it using to decide where to make the change?
- Compare the size of the repair with the size of the gap between the complete case and pairwise matrices. Which of the two decisions matters more for this data, the choice of estimator or the choice of repair?

2 A Volatility Estimate After the Regime Changed (1.5 points)

problem2.csv holds 501 daily prices for one stock. You hold \$1,000,000 of it.

Predict. Compute arithmetic returns and remove the sample mean. Plot the returns.

- Before computing any VaR, rank these five one day VaR estimates from smallest to largest, and say why you expect that order.

1. Normal, with the equally weighted standard deviation.
 2. Normal, with an exponentially weighted standard deviation at $\lambda = 0.97$.
 3. Normal, with an exponentially weighted standard deviation at $\lambda = 0.94$.
 4. A Student's t fitted by maximum likelihood.
 5. Historical simulation over the full sample.
- b. For $\lambda = 0.94$ and $\lambda = 0.97$, compute the effective sample size and the half life. Estimate from your plot how many recent days look different from the rest, and compute the fraction of the total weight each λ places on those days.
 - c. Compute the first four moments of the full sample. Predict whether a single regime of normally distributed returns could produce them.

Fit.

- d. Compute the five VaR estimates in dollars. Report the fitted t parameters.

Reconcile.

- e. Compare the result with your ranking. Explain any estimate that landed somewhere you did not expect.
- f. Compute the standard deviation of the returns inside each regime you identified in part (b). Use those to explain the excess kurtosis you found in part (c). What is the fitted t actually describing?
- g. Week 3 gives the standard error of an exponentially weighted volatility estimate as roughly $\sigma/\sqrt{2n_{\text{eff}}}$. Is the gap between the two exponentially weighted VaRs larger than that noise? Which λ would you use for a trading limit, and which for a capital number? Label your answer as an opinion where it is one.

3 When Diversification Raises VaR (1.5 points)

problem3.csv holds 10,000 simulated one year return scenarios for two bonds, A and B. Each bond is bought at 90 and pays 100 at the end of the year unless it defaults. You have \$2,000,000 to invest, and you are choosing between putting all of it in A and putting \$1,000,000 in each.

Predict. Plot the P&L distribution of both choices.

- a. Compute the first four moments of each bond. What do they tell you about a normal VaR on these positions?
- b. Count the scenarios in which each bond loses more than 20%, and the scenarios in which at least one of the two does. Before computing any risk measure, predict whether the diversified choice has a larger or a smaller VaR than the concentrated one, and the same for ES. Use the counts to justify both answers.

Fit.

- c. Compute historical VaR and ES for \$1,000,000 in A, \$1,000,000 in B, \$2,000,000 in A, and \$1,000,000 in each.
- d. Compute a normal VaR for the same positions, from the sample mean and standard deviation of each P&L.

- e. Repeat part (c) at $\alpha = 1\%$.

Reconcile.

- f. Check subadditivity for VaR and for ES at 5%, using the two \$1,000,000 positions and their combination. Explain the mechanism behind the result.
- g. The normal VaR says diversification helps. Which choice does it favour, and is it right for the right reason?
- h. At 1% the VaR result changes. Explain why, and what that says about when VaR's failure shows up.

4 Gaussian or t Copula (2 points)

problem4.csv holds 1,000 days of returns for three assets, X1 through X3. You hold \$1,000,000 of each.

Predict.

- a. Compute the first four moments of each series. From them alone, which distribution would you fit to each margin? If one number looks out of place, check how much it depends on a single observation.
- b. Convert each series to ranks scaled into $(0, 1)$ and plot every pair. Then count, for each pair, the days both series sit in their own worst 2.5%, and separately the days both sit in their own best 2.5%. How many days would you expect under independence? Do the plots and the counts suggest the joint tails are more connected than the middle of the distribution? Predict which copula will win.

Fit.

- c. Fit a normal and a t to each margin and choose between them with AICc.
- d. Transform each series to uniforms through its fitted margin. Estimate R from Kendall's τ and fit a Gaussian and a t copula. Report the log likelihood, $\hat{\nu}$, AICc, and BIC for each.
- e. Simulate 100,000 days from each copula through the fitted margins. Report VaR and ES at 5% and at 1% for the portfolio under each copula, and historically.
- f. From each simulation, compute the number of joint tail days from part (b) the copula implies over 1,000 days.

Reconcile.

- g. Which copula wins, and by how much? Compare the implied joint tail counts from part (f) with the data.
- h. Switching copulas moves one of the four portfolio risk numbers much more than the others. Which one, and why does the 5% VaR barely move?
- i. Compute each day's contribution to $\ell_t - \ell_{\text{Gauss}}$. How much of the total comes from days where no series sits in its outer 5% on either side? What does that tell you about what the information criterion is scoring?
- j. Compute the tail dependence coefficient for the most correlated pair under the fitted t copula. What is it under the Gaussian?

5 Model Based Simulation and Residual Correlation (1.5 points)

problem5.csv holds 501 daily prices for a market index, MKT, and two stocks, A and B. Assume zero expected returns. You are asked for the VaR of two portfolios:

- P1: long \$1,000,000 of A and long \$1,000,000 of B.
- P2: long \$1,000,000 of A and short \$1,000,000 of B.

We model each stock's arithmetic return as $r_i = \alpha_i + \beta_i r_{MKT} + \epsilon_i$, with the market and the errors normally distributed.

Predict.

- a. A common shortcut treats ϵ_A and ϵ_B as independent of each other. If the two stocks share an industry the market does not capture, is that shortcut going to overstate or understate the VaR of P1? Of P2? Answer with the portfolio variance formula from Week 2, before fitting anything.

Fit.

- b. Compute returns and run both regressions. Report α , β , the residual standard deviations, and the correlation between the residuals.
- c. Simulate 100,000 days of the market and the two errors, once with the full residual covariance matrix and once with its off diagonal set to zero. Build the stock returns through the fitted betas and report the VaR of both portfolios under both assumptions.
- d. Compute a delta normal VaR for both portfolios directly from the sample covariance matrix of the two stock returns.

Reconcile.

- e. Did the shortcut move each portfolio in the direction you predicted? Which portfolio is more exposed to the assumption, and why?
- f. Compare the full residual covariance VaR with the delta normal VaR from part (d). Explain the result using what Week 5 says about OLS and the joint normal. When would the two stop agreeing?

6 Code (2 points)

Your code is graded on whether we can run it and whether we can follow it. Include a README with the commands to reproduce every number in your write up. Comment the non-obvious steps, particularly anywhere you chose one convention over another.